METRIC WEDGES OF COMPACT PURELY 1-UNRECTIFIABLE SPACES HAVE LOCALLY FLAT UNIFORM SEPARATION

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Status

Result type: partial result. Status: likely valid pending human review.

This packet addresses Question 2.7 (TeX label `q:spu p1u`) in Aliaga's exposition [1]. The question asks whether, for a purely 1-unrectifiable metric space M, the algebra $\mathrm{lip}(M)$ of locally flat Lipschitz functions separates points of M uniformly. The packet proves a positive answer for metric wedges of compact purely 1-unrectifiable pointed spaces. This is not a full solution of the source question.

Open-problem evidence

The source question appears on page 6 of the rendered source PDF:

> This ends the proof. □
>
> Whether the converse of Proposition 2.6 holds is, to the best of our knowledge, an open question at the time of writing.
>
> **Question 2.7.** *Let M be a purely 1-unrectifiable metric space. Does $\mathrm{lip}(M)$ separate points of M uniformly?*

Transcription: Let M be a purely 1-unrectifiable metric space. Does $\mathrm{lip}(M)$ separate points of M uniformly?

Definitions

A real-valued Lipschitz function f on a metric space M is locally flat if for every $p \in M$,

$$\lim_{r \downarrow 0} \sup \left\{ \frac{|f(u) - f(v)|}{d(u,v)} : 0 < d(u,v),\ d(u,p) < r,\ d(v,p) < r \right\} = 0.$$

The space $\mathrm{lip}(M)$ consists of locally flat Lipschitz functions. It separates points uniformly if for every $x, y \in M$ and every $\varepsilon > 0$ there is $f \in \mathrm{lip}(M)$ with $\mathrm{Lip}(f) \leq 1$ and

$$f(y) - f(x) > d(x,y) - \varepsilon.$$

Let $\{(K_i, o_i)\}_{i \in I}$ be pointed metric spaces. Their metric wedge $W = \bigvee_{i \in I} K_i$ is obtained by identifying all basepoints o_i to a single point o. If a, b lie in the same branch K_i, their distance in W is $d_i(a,b)$. If $a \in K_i$, $b \in K_j$, $i \neq j$, then

$$d_W(a,b) = d_i(a, o_i) + d_j(b, o_j).$$

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Proof intuition

For a compact purely 1-unrectifiable branch, the AGPP theorem [2] gives a locally flat, 1-Lipschitz function that almost realizes the distance between any two points. If such a branch function is normalized to vanish at the wedge basepoint, extending it by zero to all other branches stays 1-Lipschitz because the wedge distance to any other branch includes the whole distance back to the basepoint. For points in two different branches, use one normalized separator on each branch, with opposite signs. The only point at which local flatness could fail is the common basepoint; it does not fail because each branch function is $o(d(\cdot, o))$ near the base, so for two nearby points in distinct branches the numerator is bounded by the sum of two small radial errors.

The original question remains open outside this wedge class.

Main theorem

Theorem 1. *Let $\{(K_i, o_i)\}_{i\in I}$ be any family of pointed compact metric spaces. Assume every K_i is purely 1-unrectifiable. Then*

$$W = \bigvee_{i\in I} K_i$$

has the following property: $\mathrm{lip}(W)$ *separates points of* W *uniformly.*

Proof. We use the compact theorem of Aliaga, Gartland, Petitjean, and Prochazka: if K is compact and purely 1-unrectifiable, then $\mathrm{lip}(K)$ separates points of K uniformly [1, Theorem 2.10], [2, Theorem A].

Fix $x, y \in W$ and $\varepsilon > 0$. If $x = y$, the zero function works, so assume $x \neq y$.

First suppose that x, y belong to the same branch K_i, where the basepoint o is identified with o_i if one of the points is the wedge point. By the compact theorem there is $h \in \mathrm{lip}(K_i)$ such that $\mathrm{Lip}(h) \le 1$ and

$$h(y) - h(x) > d_i(x, y) - \varepsilon.$$

Replacing h by $h - h(o_i)$, assume $h(o_i) = 0$. Define $F : W \to \mathbb{R}$ by $F = h$ on K_i and $F = 0$ on all other branches. The function is 1-Lipschitz: inside K_i this is clear; if $a \in K_i$ and b is in another branch, then

$$|F(a) - F(b)| = |h(a)| \le d_i(a, o_i) \le d_W(a, b);$$

and if both points are outside K_i, the values are both zero. Thus $F(y) - F(x) > d_W(x, y) - \varepsilon$.

It remains in this case only to verify local flatness. Away from the wedge point, a small ball is contained in a single branch, so local flatness follows from that of h or from the fact that F is locally constant. At the wedge point, let $\eta > 0$. Since $h \in \mathrm{lip}(K_i)$ and $h(o_i) = 0$, there is $\delta > 0$ such that $|h(a)| \le \eta d_i(a, o_i)$ whenever $a \in K_i$ and $d_i(a, o_i) < \delta$, and also such that $|h(a) - h(b)| \le \eta d_i(a, b)$ for $a, b \in K_i$ within δ of o_i. For any $u, v \in W$ within δ of o, if both are in K_i the latter estimate applies; if exactly one is in K_i, say $u \in K_i$, then

$$|F(u) - F(v)| = |h(u)| \le \eta d_i(u, o_i) \le \eta d_W(u, v);$$

and if neither is in K_i, the numerator is zero. Hence F is locally flat at o.

Now suppose $x \in K_i \setminus \{o\}$ and $y \in K_j \setminus \{o\}$ with $i \neq j$. Put $r_x = d_i(x, o_i)$ and $r_y = d_j(y, o_j)$. By the compact theorem choose $h_i \in \mathrm{lip}(K_i)$ and $h_j \in \mathrm{lip}(K_j)$ with Lipschitz constants at most 1, normalized by $h_i(o_i) = h_j(o_j) = 0$, such that

$$h_i(x) > r_x - \varepsilon/2, \qquad h_j(y) > r_y - \varepsilon/2.$$

Define $F : W \to \mathbb{R}$ by $F = -h_i$ on K_i, $F = h_j$ on K_j, and $F = 0$ on all other branches. Then

$$F(y) - F(x) = h_j(y) + h_i(x) > r_y + r_x - \varepsilon = d_W(x, y) - \varepsilon.$$

The same radial estimates prove that F is 1-Lipschitz. Inside a selected branch this is the Lipschitz bound for h_i or h_j. Between the two selected branches,

$$|F(a) - F(b)| \le |h_i(a)| + |h_j(b)| \le d_i(a, o_i) + d_j(b, o_j) = d_W(a, b),$$

and the estimates against an unselected branch are the same with one term missing. Local flatness away from o is branch-local. At o, choose δ so that both selected functions have the corresponding radial and same-branch local-flat estimates with error η. If two nearby points lie in different selected branches, the preceding display improves to $|F(u) - F(v)| \le \eta(d_W(u, o) + d_W(v, o)) = \eta d_W(u, v)$; all other cases are easier. Hence $F \in \mathrm{lip}(W)$.

This proves uniform separation for all pairs of points in W. □

Lemma 1. *If each branch K_i is complete, then the metric wedge W is complete. In particular, the wedges in Theorem 1 are complete.*

Proof. Let (z_n) be a Cauchy sequence in W and put $r_n = d_W(z_n, o)$. If $r_n \to 0$, then $z_n \to o$. Otherwise some tail has points with radius bounded below. Choose $\alpha > 0$ and infinitely many n with $r_n > 2\alpha$. Since (z_n) is Cauchy, there is N such that $d_W(z_m, z_n) < \alpha$ for $m, n \ge N$. Taking one $n \ge N$ with $r_n > 2\alpha$, every later point must lie in the same branch as z_n; a point in another branch, including the base, would be at distance at least $r_n > \alpha$ from z_n. The tail is therefore Cauchy in one complete branch and converges there. □

Corollary 1. *The source question has an affirmative answer for every complete metric wedge of compact purely 1-unrectifiable pointed metric spaces. Such wedges may be noncompact, for instance when there are infinitely many nontrivial branches with diameters bounded away from zero.*

Proof. The affirmative answer is Theorem 1, and completeness is Lemma 1. Noncompactness follows, for example, by taking infinitely many two-point branches of the same positive length. □

Verification notes

The proof has no computational dependency. The key checks are:

- Normalizing each branch function by $h(o_i) = 0$ gives $|h(a)| \le d_i(a, o_i)$, so extension by zero is 1-Lipschitz across branches.
- In the two-branch case the signs make $F(y) - F(x)$ add the two radial branch distances, exactly the wedge distance.
- Local flatness at the wedge point follows from radial smallness $|h(a)| = o(d_i(a, o_i))$ for each selected branch.

Novelty and limitations

The local run indexes were searched for `2606.02918`, `q:spu`, `locally flat`, `separates points uniformly`, `metric wedge`, and `purely 1-unrectifiable`. No duplicate packet for this question or wedge closure result was found. A bounded web search on 2026-06-28 for exact phrases including "lip(M) separates points", "locally flat Lipschitz functions" with "metric wedge", and "purely 1-unrectifiable" with "wedge" found the AGPP compact theorem but no exact wedge result.

This packet does not solve Question 2.7 for arbitrary purely 1-unrectifiable metric spaces. It proves a closure theorem that manufactures possibly noncompact positive examples from compact positive ones.

Human review recommendation

Please check the basepoint local-flatness argument and the completeness lemma. The rest of the proof is a direct branchwise use of the compact theorem and the defining formula for the wedge metric.

References

[1] R. J. Aliaga, *Lipschitz-free spaces and purely 1-unrectifiable metric spaces*, arXiv:2606.02918, 2026. https://arxiv.org/abs/2606.02918

[2] R. J. Aliaga, C. Gartland, C. Petitjean, and A. Prochazka, *Purely 1-unrectifiable metric spaces and locally flat Lipschitz functions*, Trans. Amer. Math. Soc. **375** (2022), 3529–3567. https://arxiv.org/abs/2103.09370